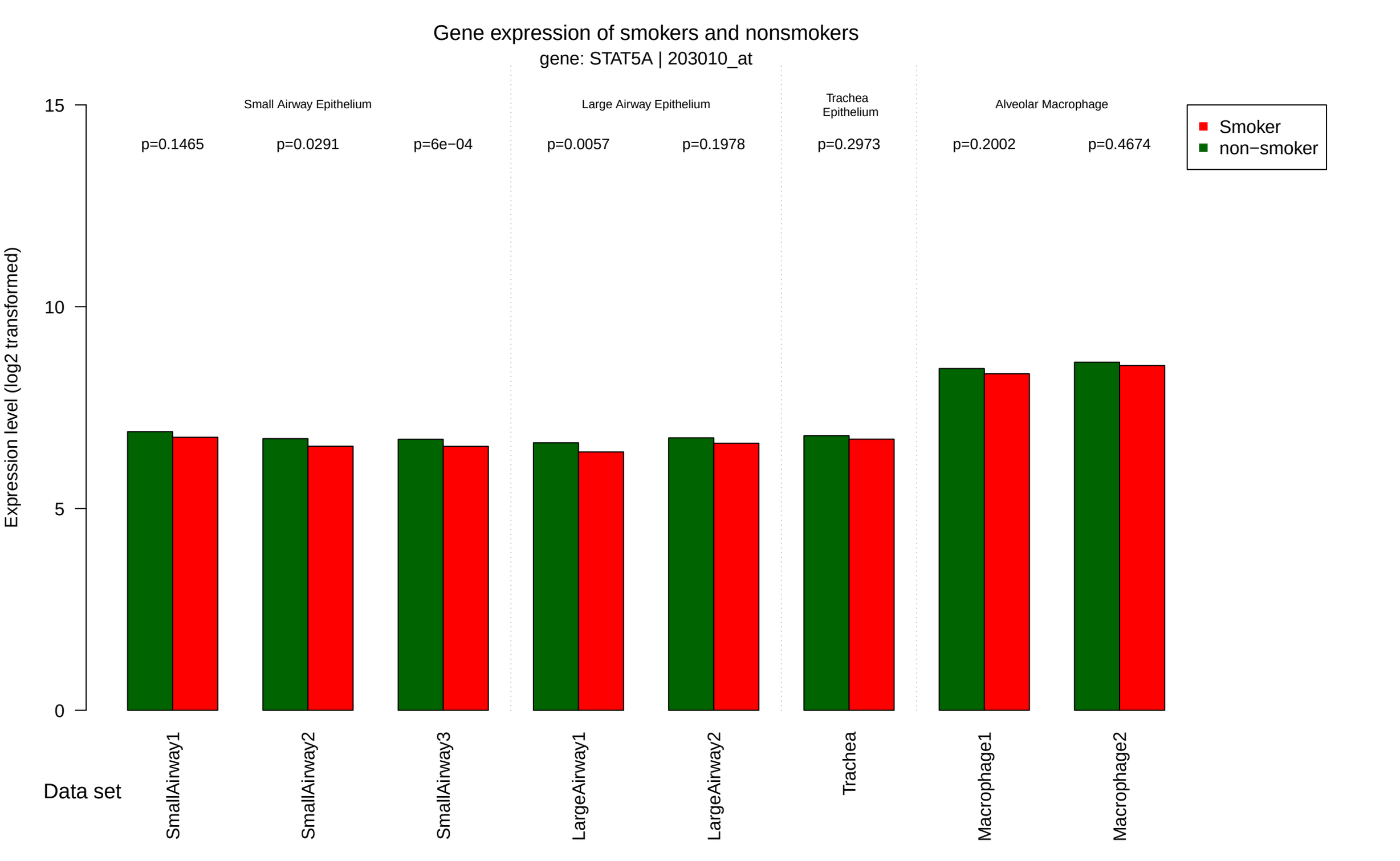
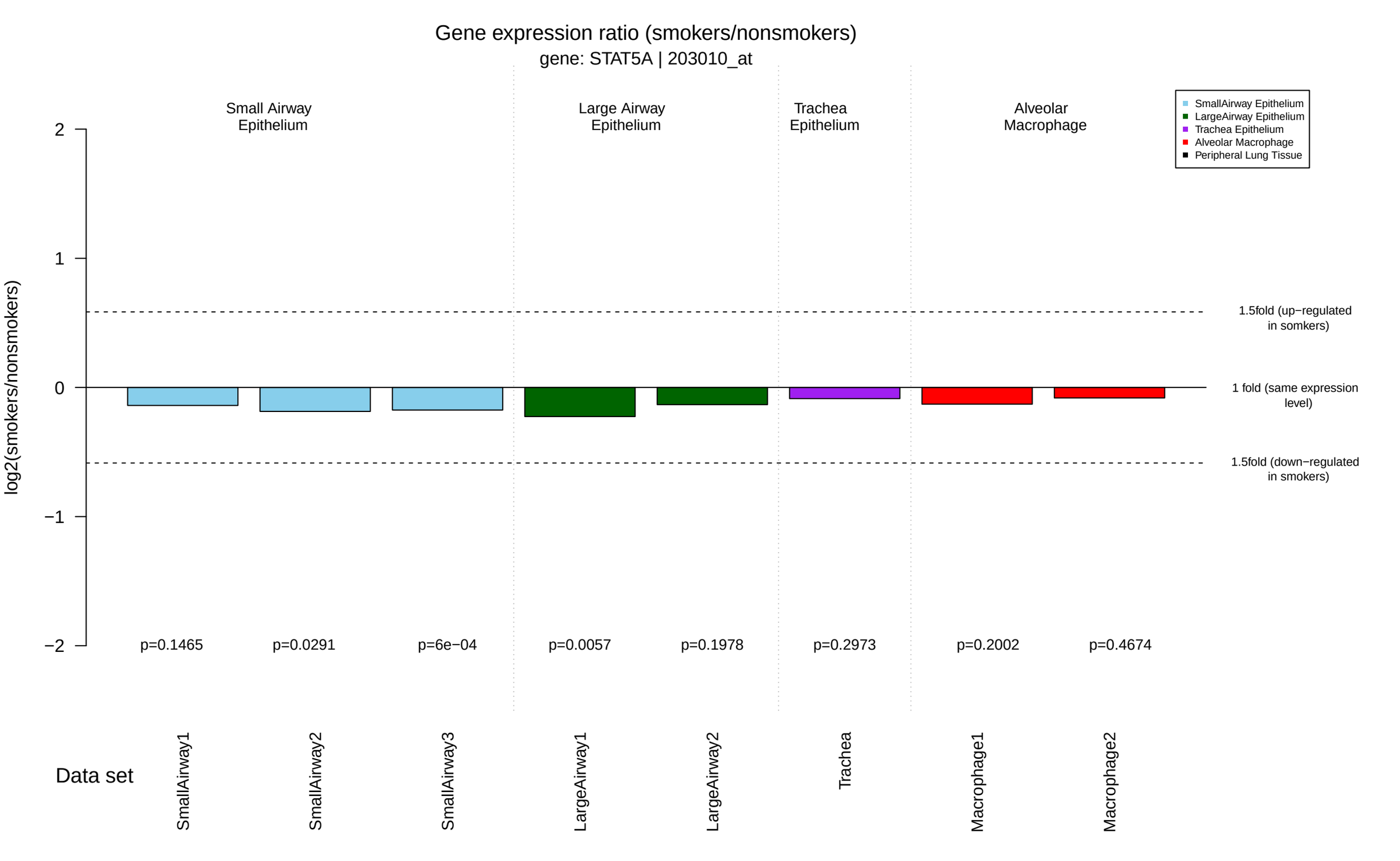
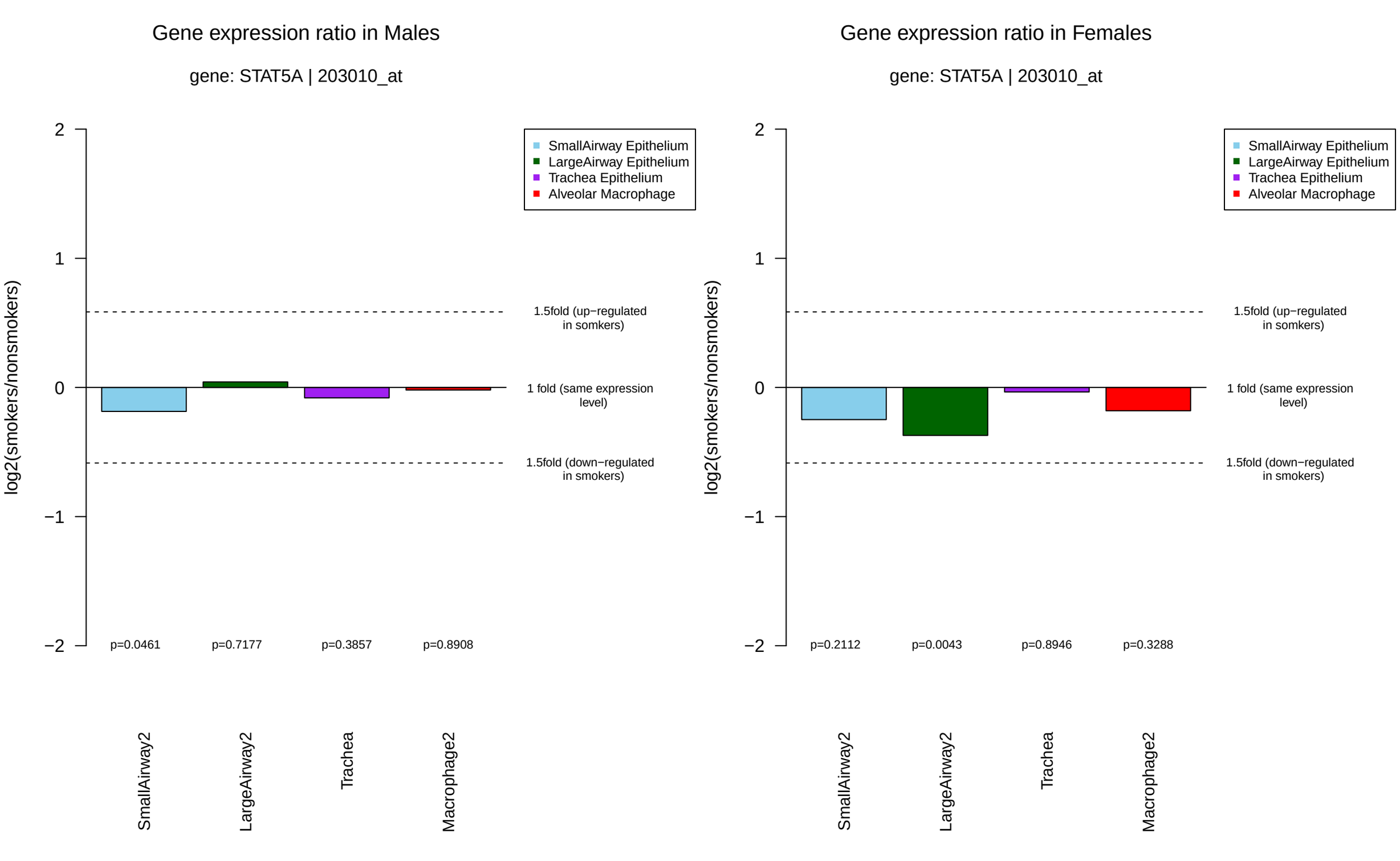
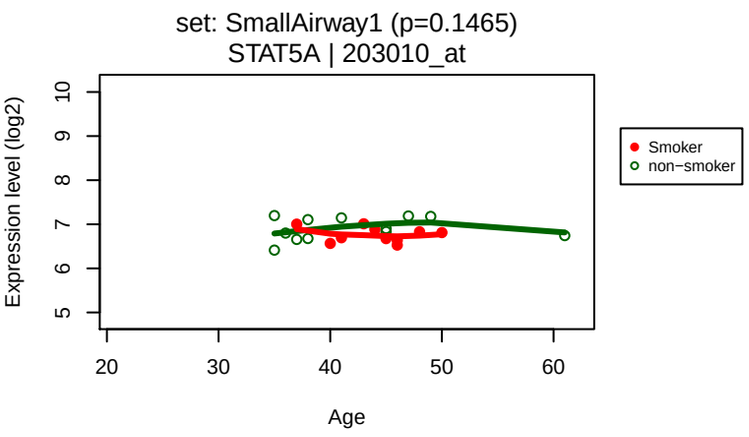
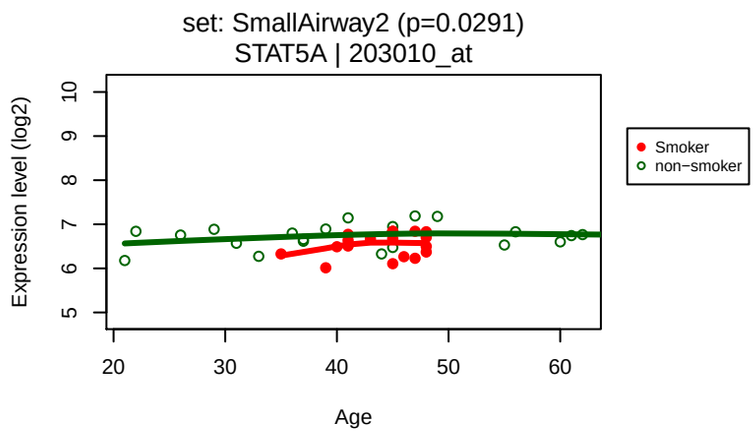
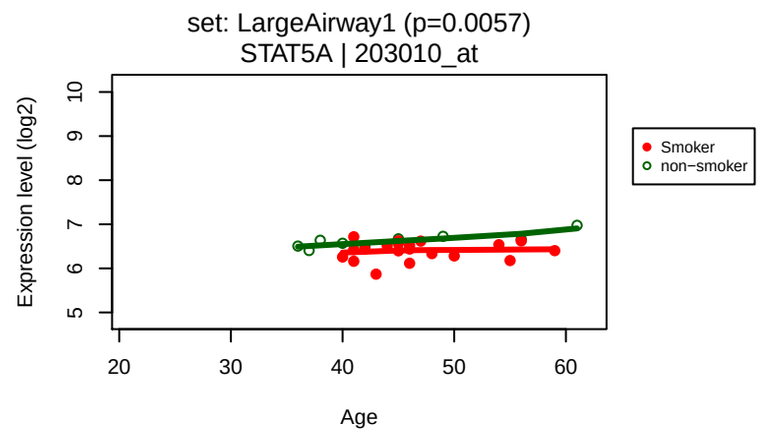

Gene expression of smokers and nonsmokers
gene: STAT5A | 203010_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1465
p=0.0291
p=6e−04
p=0.0057
p=0.1978
p=0.2973
p=0.2002
p=0.4674
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

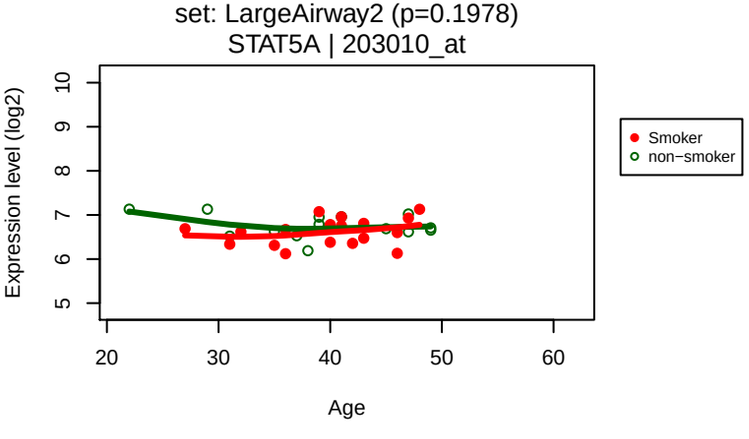

Gene expression ratio (smokers/nonsmokers)
gene: STAT5A | 203010_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1465
p=0.0291
p=6e−04
p=0.0057
p=0.1978
p=0.2973
p=0.2002
p=0.4674
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

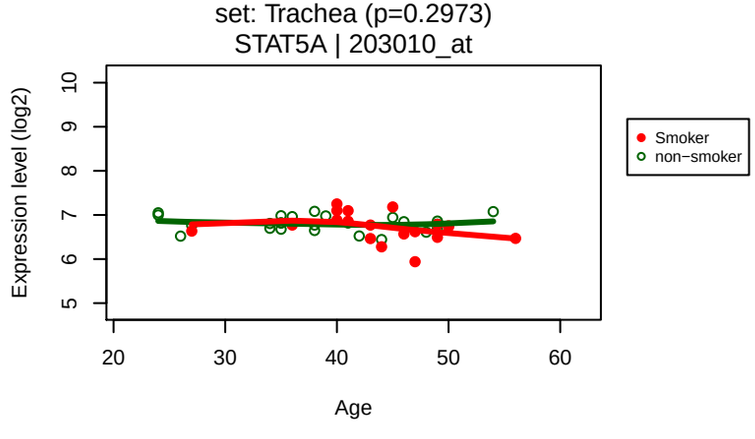

Gene expression ratio in Males
gene: STAT5A | 203010_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0461
p=0.7177
p=0.3857
p=0.8908
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAT5A | 203010_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2112
p=0.0043
p=0.8946
p=0.3288
SmallAirway2
LargeAirway2
Trachea
Macrophage2

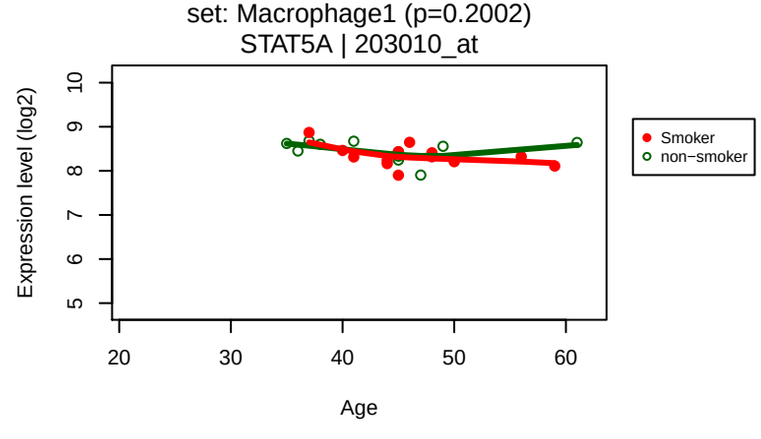

set: SmallAirway1 (p=0.1465)
STAT5A | 203010_at
Expression level (log2)
Age
Smoker
non−smoker

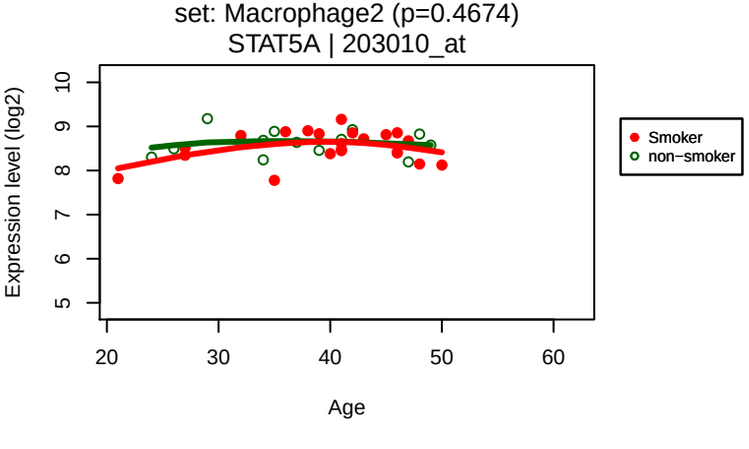

set: SmallAirway2 (p=0.0291)
STAT5A | 203010_at
Expression level (log2)
Age
Smoker
non−smoker

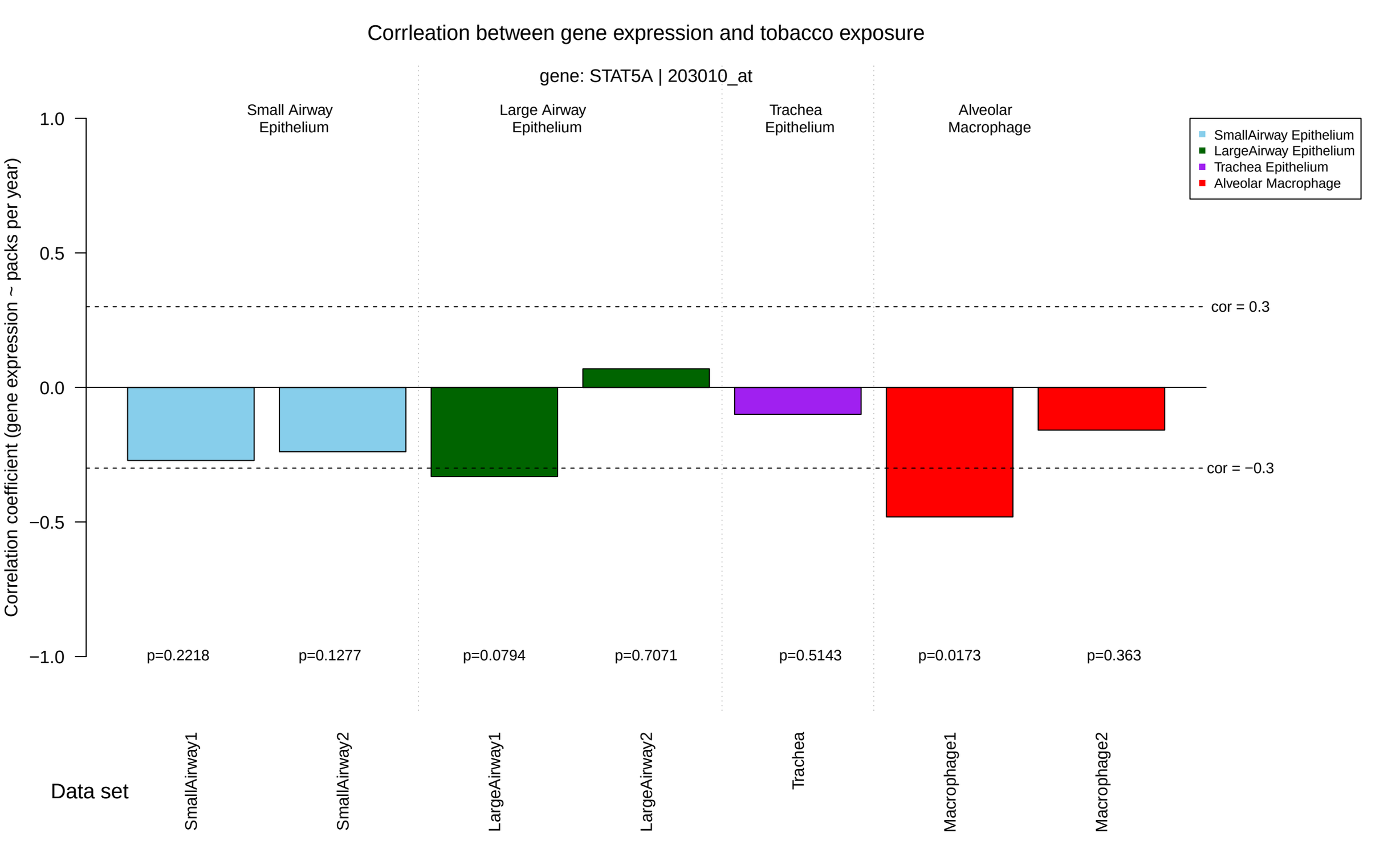

set: LargeAirway1 (p=0.0057)
STAT5A | 203010_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1978)
STAT5A | 203010_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2973)
STAT5A | 203010_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2002)
STAT5A | 203010_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4674)
STAT5A | 203010_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAT5A | 203010_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2218
p=0.1277
p=0.0794
p=0.7071
p=0.5143
p=0.0173
p=0.363
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2